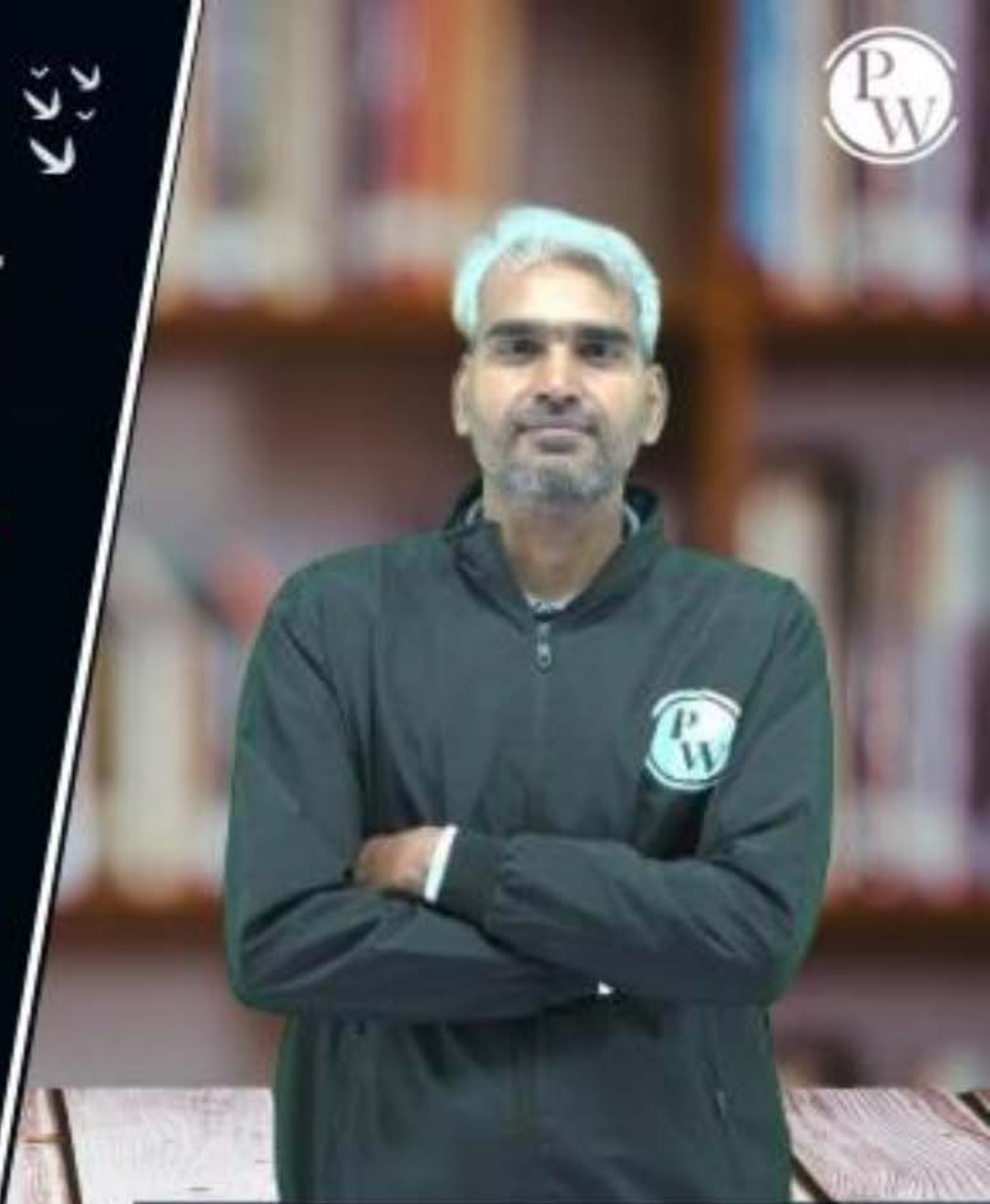
# Computer Science & IT

## Discrete Mathematics

**Graph Theory**

**Lecture No. 12**



By- Vishal Sir

# Recap of Previous Lecture

- **Topic** Line / Edge Covering
- **Topic** Minimal & Minimum line covering
- **Topic** Line covering number
- **Topic** Line independent set
- **Topic** Maximal & Maximum line independent set

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# Topics to be Covered

- **Topic** Vertex Covering
- **Topic** Minimal & Minimum vertex covering
- **Topic** Vertex independent set
- **Topic** Maximal & Maximum vertex independent set
- **Topic** Connectivity

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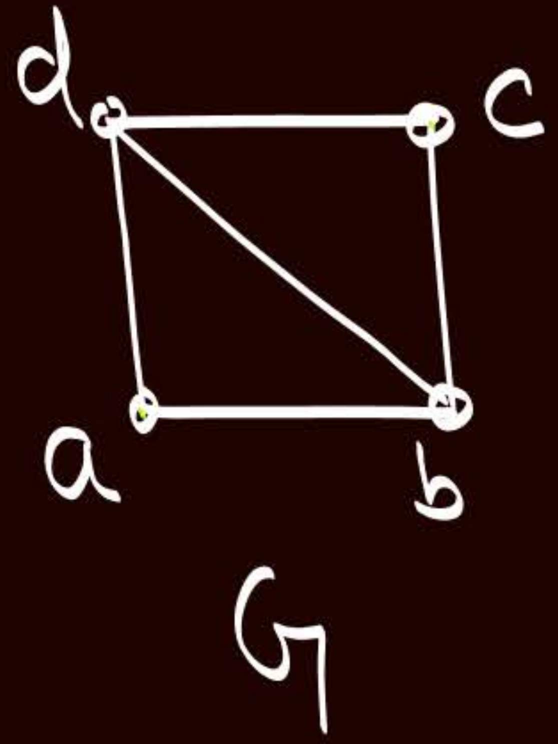
## Topic : Vertex Covering

- Let G=(V,E) be a graph.

A subset K of set of vertices V is called vertex covering of graph G if every edge of the graph is incident with at least one vertex of set K {i.e. Every edge of graph G must have at least one of its "end vertex" in subset K}

eg:




$K_1 = \{a, b, c, d\}$

✓ $K_2 = \{a, b, c\}$

✓ $K_3 = \{b, d\}$

✓ $K_4 = \{a, c, d\}$

$K_5 = \{a, b, d\}$

$K_1, K_2, K_3, K_4, K_5, \ldots$ are vertex Covering of graph G

## Topic : Minimal vertex covering

- A vertex covering of graph G is called minimal vertex covering if no vertex can be deleted from the set without destroying its ability to cover all the edges.

In the above example,

$K_2$, $K_3$ are $K_4$ are minimal vertex covering

# Topic : Minimum vertex covering

/ Smallest Minimal Vertex Covering

* Vertex coverings of graph G with minimum number of vertices are called minimum vertex coverings of graph G

In the above example,

$K_3$ is the minimum vertex covering

* Every minimum vertex Covering is also a minimal Vertex Covering but Every minimal vertex covering need not be a minimum vertex covering

# Topic : Vertex covering number ($\alpha_2$)

Vertex Covering number of graph G ($\alpha_2$) = No. of vertices in any one of the minimum vertex Covering of G

In the above example,

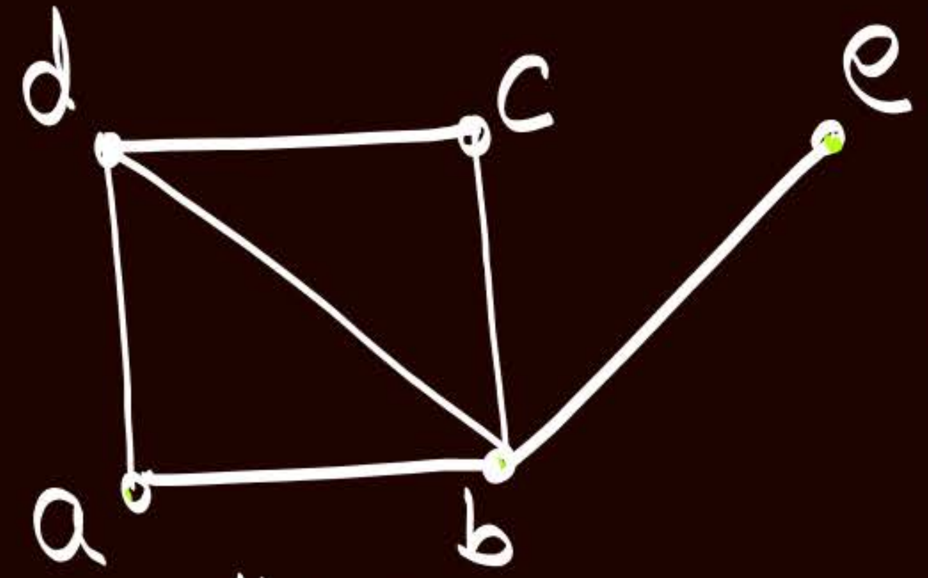
$$\alpha_2 = 2$$

w.r.t. '$K_3$'

## Topic : Vertex independent set

- Let $G = (V, E)$ be a graph.
- A subset 'S' of set of vertices V is called a vertex independent set if no two vertices of subset S are adjacent to each other.

eg:




"G"

$S_1 = \{a\}$

{ A set containing a single vertex will always be a vertex independent set }

$S_2 = \{a, c\}$

✓ $S_3 = \{a, c, e\}$

✓ $S_4 = \{b\}$

✓ $S_5 = \{d, e\}$

$S_6 = \{c, e\}$

All subsets $S_1, S_2, S_3, S_4, S_5, S_6, \ldots$ are vertex independent set w.r.t. graph G.

# Topic : Maximal vertex independent set

- A vertex independent set of graph G is called a maximal vertex independence set if no new vertex of the graph can be added to the set without destroying its property of being independent.

In the above example.

$S_3$, $S_4$ & $S_5$ are maximal vertex independent set.

# Topic : Maximum vertex independent set

~~Longest~~ Maximal vertex independent ~~set~~

- Vertex independent sets with maximum number of vertices are called maximum vertex independent set of graph $G$

In the above example.

$S_3$ is the maximum vertex independent set.

# Topic : Vertex independence number ($\beta_2$)

Vertex independen Number ($\beta_2$) of graph G = No. of vertices in any one of the maximum vertex independent set of graph G.

In the above example,

$\beta_2 = 3$

w.r.t. '$S_3$'

* Every maximum vertex independent set is also a maximal vertex independent set, but Every maximal vertex independent set need not be a maximum vertex independent set.

## Topic : NOTE

For any graph $G = (V, E)$

$$\alpha_2 + \beta_2 = |V(G)|$$

# Topic : NOTE

* For a graph $G = (V, E)$

Note: Similar statement need not hold true for line covering & line independent set

(1) If 'S' is a vertex independent set of graph G, then 'V-S' is a vertex covering of graph G.

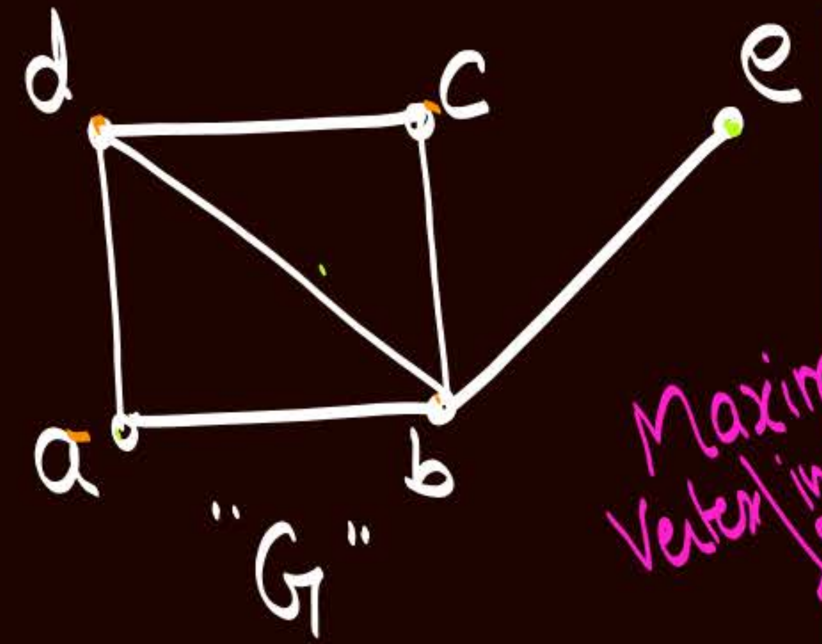
{ if S is maximal then 'V-S' is minimal
if S is maximum then 'V-S' is minimum }

and

(2) If 'K' is a vertex covering of graph 'G', then 'V-K' is a vertex independent set of graph G.

{ if 'K' is minimal, then 'V-K' is maximal
& if 'K' is minimum, then 'V-K' is maximum }

eg:




$V = \{a, b, c, d, e\}$

## 'S'

"vertex independent set"

$S_1 = \{a\}$

$S_2 = \{a, c\}$

Maximum Vertex independent set

✓ $S_3 = \{a, c, e\}$

✓ $S_4 = \{b\}$

✓ $S_5 = \{d, e\}$

$S_6 = \{c, e\}$

Maximal Vertex Independent Set

## 'V−S'

"vertex Covering"

$\{b, c, d, e\}$

$\{b, d, e\}$

$\{b, d\}$

Minimum Vertex Covering

$\{a, c, d, e\}$

$\{a, b, c\}$

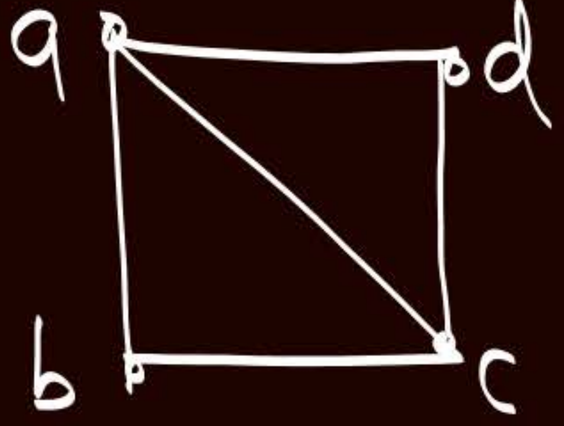
$\{a, b, d\}$

Minimal Vertex Covering

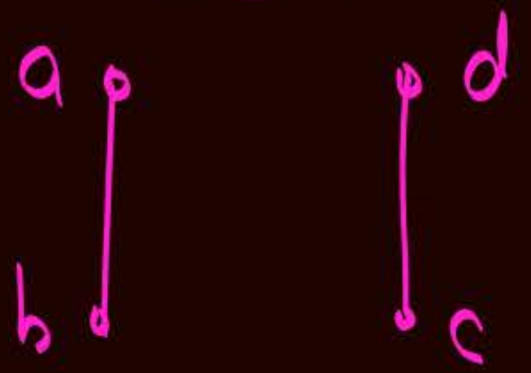
Note:

Let $G = (V, E)$ be a graph,
if '$L$' is a line covering of graph $G$, then
"$E - L$" need not be a line independent set.

eg Let

$G =$ 

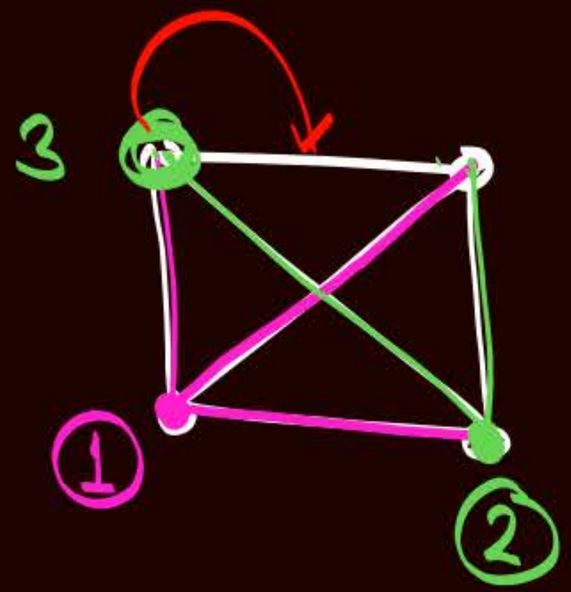
$L_1 = \{\{a, b\}, \{c, d\}\}$



'$L_1$' is line covering of graph $G$.

but
$E - L_1 = \{\{a, d\}, \{a, c\}, \{b, c\}\}$
is not a line independent set

Q: Find $\alpha_2$ and $\beta_2$ for Complete graph $K_n$

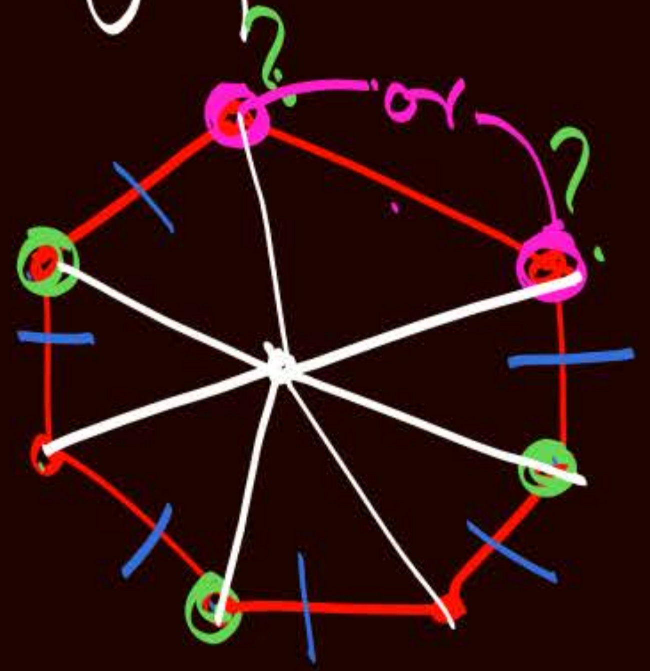


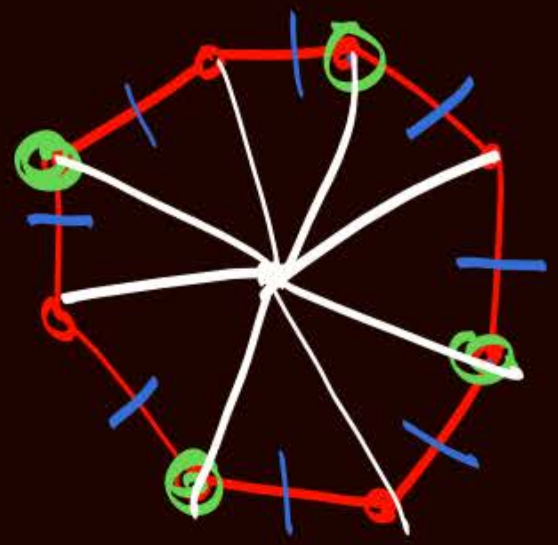

- Vertex Covering no. of $K_n$ $= (n-1)$
  $\alpha_2(K_n)$

- Vertex independence no. of $K_n$ $= 1$
  $\beta_2(K_n)$

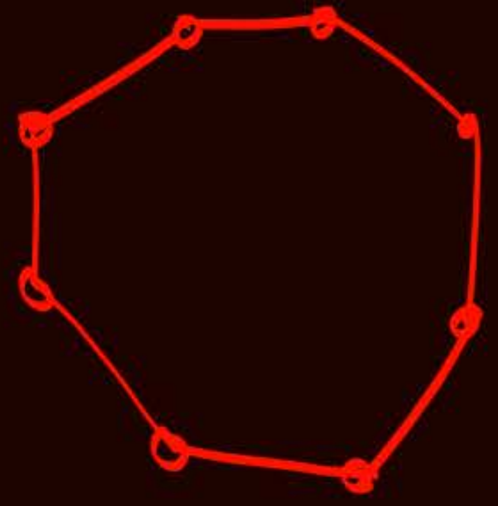
$$\alpha_2 + \beta_2 = n$$

Q: Find $\alpha_2$ and $\beta_2$ for Cycle graph $C_n$



- Vertex Covering no. of $C_n$ $= \left\lceil \frac{n}{2} \right\rceil$
  $\alpha_2(C_n)$



- Vertex independence no. of $C_n$ $= \left\lfloor \frac{n}{2} \right\rfloor$
  $\beta_2(C_n)$

$$\alpha_2 + \beta_2 = n$$

Q: Find $\alpha_2$ and $\beta_2$ for Wheel graph $W_n$

hub must be selected to cover all the spokes of wheel graph

- Vertex Covering no. of $W_n$ $\alpha_2(W_n)$ $= 1 + \left\lceil \frac{n-1}{2} \right\rceil = \left\lceil \frac{n+1}{2} \right\rceil$

To cover the edges of Cycle $C_{n-1}$

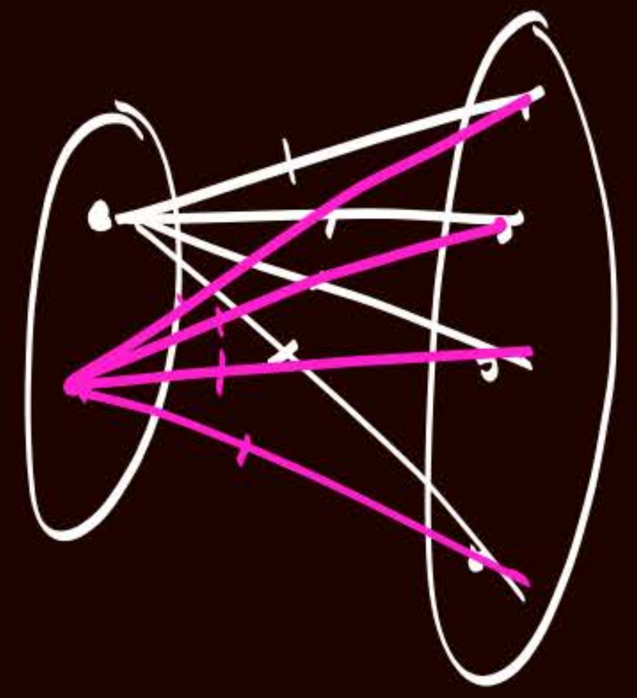
- Vertex independence no. of $W_n$ $\beta_2(W_n)$ $= \left\lfloor \frac{(n-1)}{2} \right\rfloor$

No. of vertices in largest vertex independent set

if we select the hub vertex, then we will not be able to select any other vertex because all are adjecent to hub vertex

i.e. w.r.t. largest vertex independent set we must alway avoid hub vertex, and $\therefore$ we are left with a cycle of $(n-1)$ vertices

Q: Find $\alpha_2$ and $\beta_2$ for Complete bipartite graph $K_{m,n}$



- Vertex Covering no. of $K_{m,n}$ = $\text{Min}(m,n)$
  $\alpha_2(K_{m,n})$

- Vertex independence no. of $K_{m,n}$ = $\text{Max}(m,n)$
  $\beta_2(K_{m,n})$

Q: Find $\alpha_2$ and $\beta_2$ for Star graph with 'n' vertices

- Vertex Covering no. of Star graph with 'n' vertices $= \underline{1}$

$= K_{1,n-1}$ or $K_{n-1,1}$ $= \text{Min}(1, n-1)$

- Vertex independence no. of Star graph with 'n' vertices $= (n-1)$

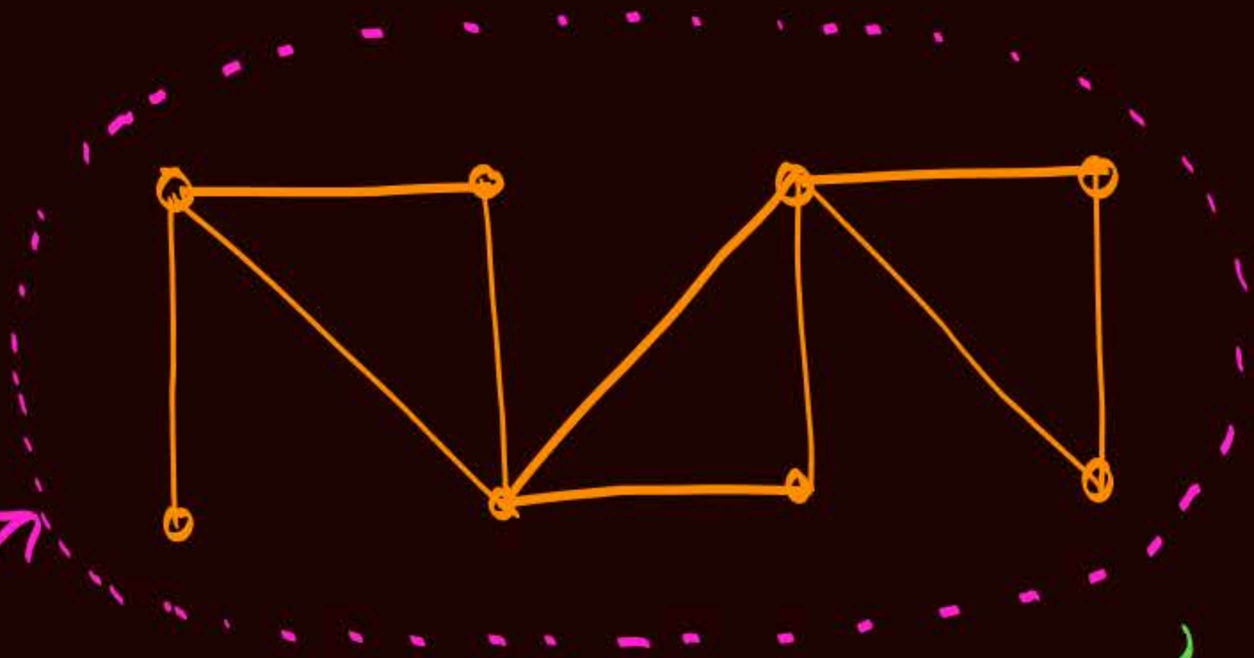
$= K_{1,n-1}$ or $K_{n-1,1}$ $= \text{Max}(1, n-1)$

$\boxed{\text{Connectivity}}$

# Topic : Connectivity

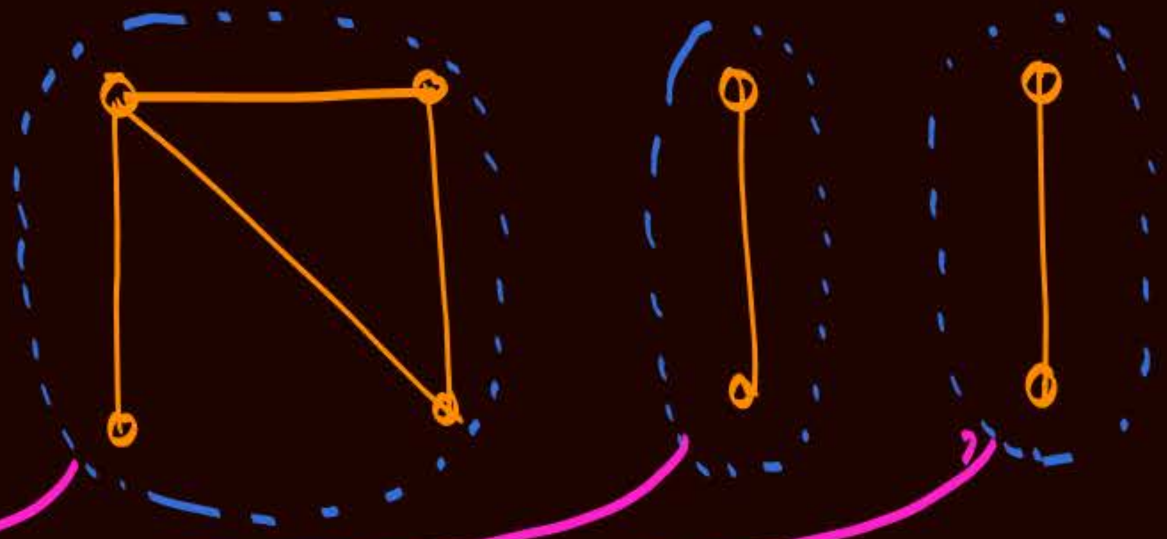
A graph is said to be connected if there exists a path between every pair of vertices.

If a graph (with 2 or more vertices) is not connected, then it will have two or more connected components.



Single Connected Component

Connected graph



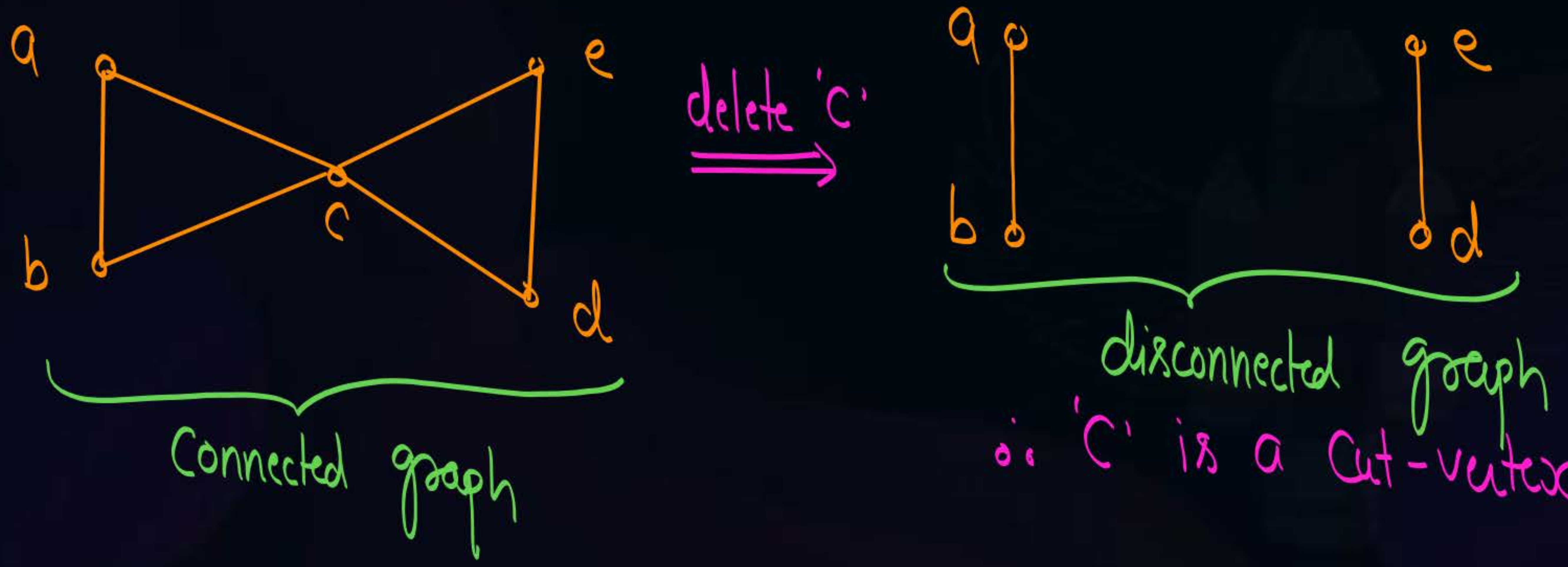
Three Connected Components

disconnected graph

## Topic : Cut-vertex

/ "Articulation point"

Let G be a connected graph, a vertex $v \in G$ is called a cut vertex if deletion of vertex "$v$" from graph G makes the graph disconnected.

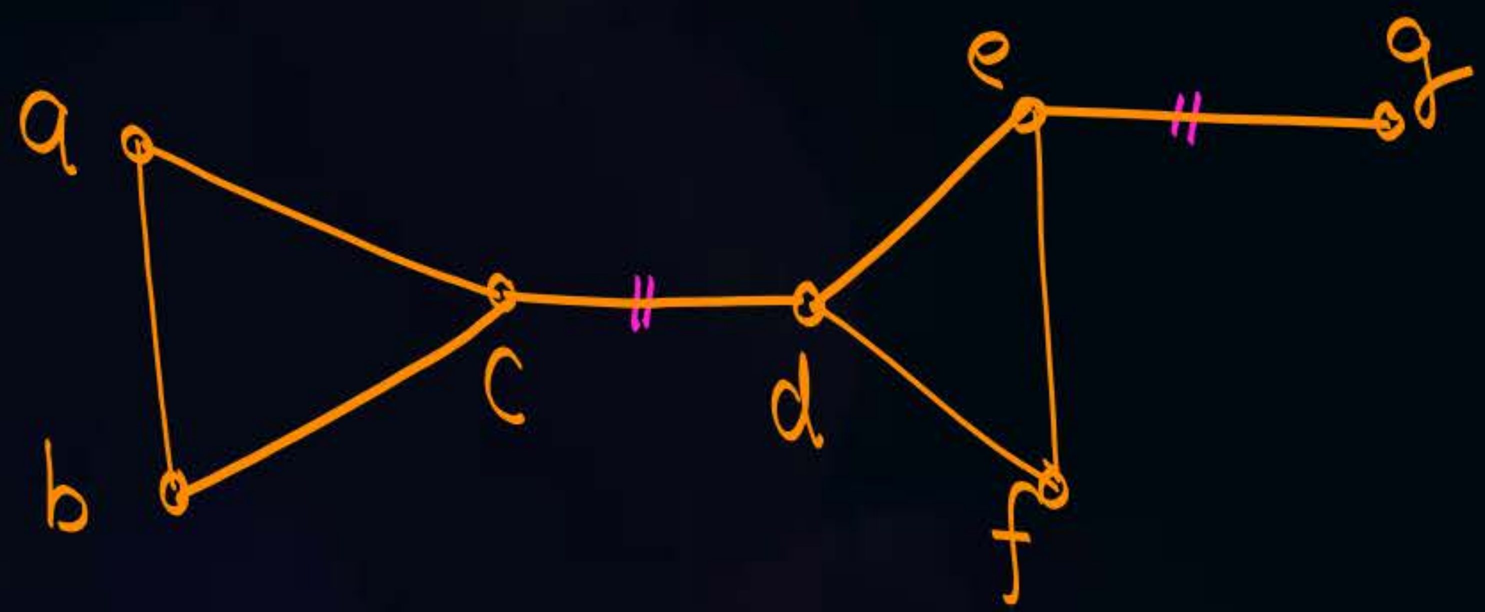



∴ 'C' is a Cut-vertex

# Topic : Cut-edge / Bridge

Let G be a connected graph, an edge $e \in G$ is called a cut edge or bridge if deletion of edge "$e$" from graph G makes the graph disconnected.

e.g.




Edge $\{c,d\}$ is a Cut-edge

Edge $\{e,g\}$ is also a Cut-edge

# Topic : NOTE

(1) An edge of a connected graph G can be a Cut-edge (bridge) if and only if that edge is not a part of any cycle in the graph.

(2) At least one vertex associated with a Cut-edge is a cut-vertex of that graph

e.g. w.r.t. Cut-edge {c,d} in above example, both c & d are Cut-vertex

w.r.t. Cut-edge {e,g} in above example, 'g' is not a cut-vertex but 'e' is a Cut-vertex

Note:- Existance of Cut-vertex in a graph does not imply the existance of cut-edge, but

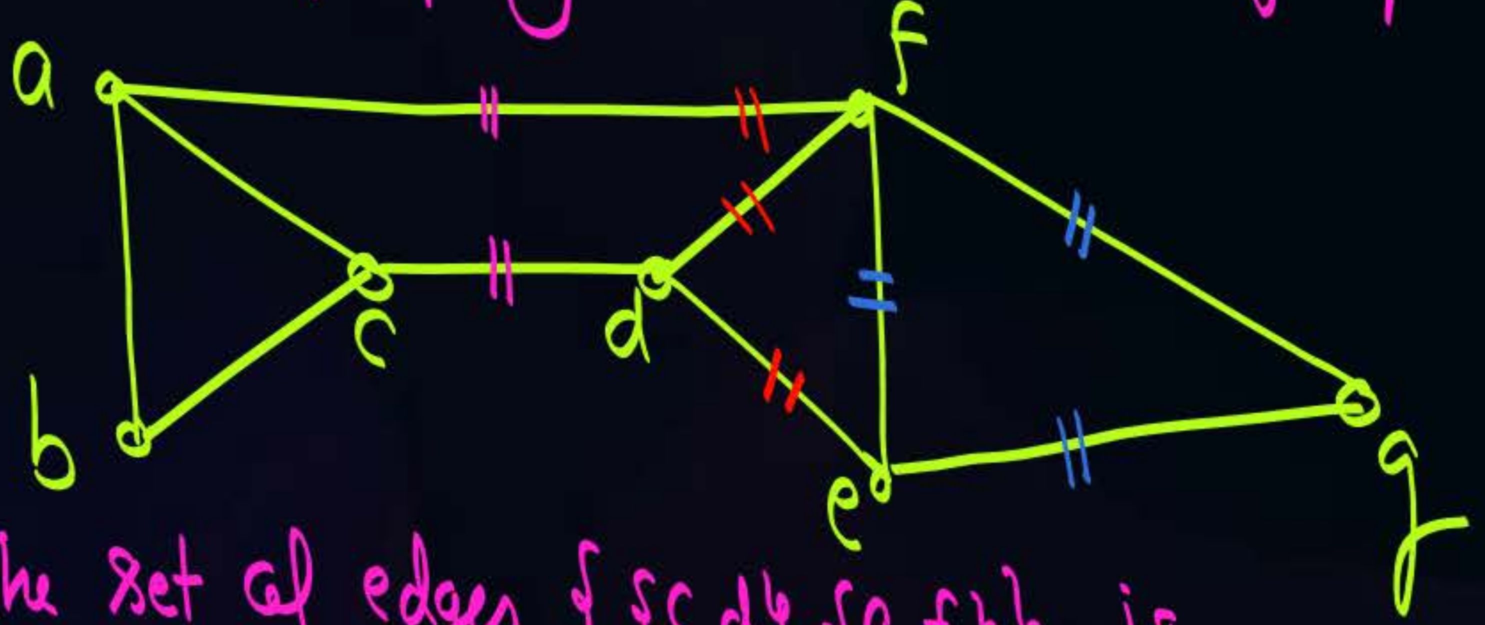
Existance of a Cut-edge in a graph implies the existance of Cut-vertex.

# Topic : Cut-set

i.e., No edge can be removed from the set.

A minimal set of edges in a connected graph G, such that removal of all the edges of that set from the graph G makes the graph disconnected is called cut-set of graph G.

if we leave any edge from the set, then it will loose its property to make the graph disconnected




The set of edges {{c,d},{a,f}} is a cut set of graph G

eg: {{d,e},{d,f},{a,f}} is also a cut-set

Note: {{e,f},{e,g},{f,g}} is not a cut-set, because it is not minimal, we can remove the edge {e,f} from the set

# Topic : Vertex connectivity

Let G be a connected graph, the minimum number of vertices whose removal from graph G makes the graph disconnected is called vertex connectivity of graph G. Vertex connectivity of a graph G is denoted by $\kappa(G)$

* Vertex connectivity of graph is 1 if and only if cut-vertex exists in the graph.

* If graph G is already a disconnected graph, then $K(G) = 0$
* If G is a connected graph and cut-vertex does not exist in graph G, then $K(G) \geq 2$

# Topic : Edge connectivity

Let G be a connected graph, the minimum number of edges whose removal from graph G makes the graph disconnected is called edge connectivity of graph G. Edge connectivity of a graph G is denoted by $\lambda(G)$

- Edge connectivity of graph is 1 if and only if cut-edge exists in the graph.

- If graph G is already a disconnected graph, then $\lambda(G) = 0$
- If graph G is a connected graph and cut-edge does not exist in graph G, then $\lambda(G) \geq 2$

Topic : NOTE

(1) For a graph $G$, $\lambda(G) \leq \delta(G)$ — eqⁿ (1)

$\delta(G)$ is minimum degree in graph $G$.

(2) For a graph $G$,

$$K(G) \leq \lambda(G) \text{ — eq}^n \text{ (2)}$$

By eqⁿ (1) and eqⁿ (2)

$$K(G) \leq \lambda(G) \leq \delta(G)$$

Q:- Find $\lambda(G)$ and $K(G)$ for Complete graph $K_n$.

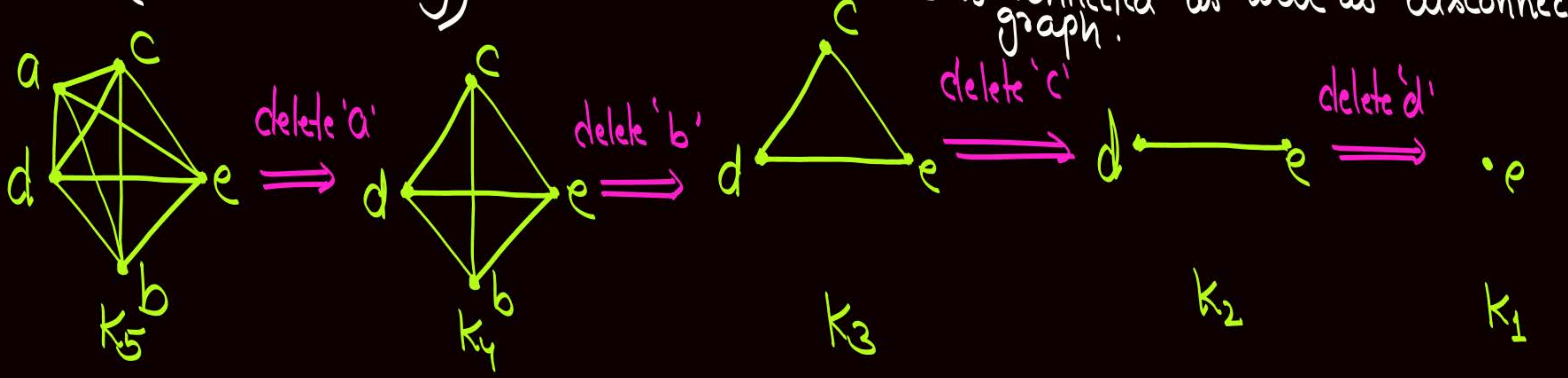
(Edge-Connectivity)

$\lambda(G)$ w.r.t. $K_n = (n-1)$

$K(G)$ w.r.t. $K_n = (n-1)$

(vertex Connectivity)

{ after deletion of $(n-1)$ vertices from $K_n$, we will get a single isolated vertex, it can be treated as connected as well as disconnected graph. }



Note: If an isolated Vertex exists along with other Components in the graph, then isolated Vertex is treated as a Connected Component

H.W. Find vertex Connectivity ($K(G)$) and Edge Connectivity ($\lambda(G)$) For the following graphs.

| | Vertex Connectivity | Edge Connectivity |
|---|---|---|
| ① Complete graph $K_n$ | $(n-1)$ | $(n-1)$ |
| ② Cycle graph $C_n$ | | |
| ③ Wheel graph $W_n$ | | |
| ④ Complete bipartite graph $K_{m,n}$ | | |
| ⑤ Star graph with $n$-vertices | | |

H.W.

#Q. If G is a forest with n vertices and k connected components, how many edges does G have?

H.W. [MCQ]

#Q. In a connected graph G if we delete an edge then no. of components are

A 0 or 1

B 1 or 2

C 2

D $\geq 2$

[MCQ]

H.W.

with 'n' vertices

#Q. In a connected graph G if we delete a vertex then number of components are

A $\leq 2$

B 1 or 2

C Lies between 1 and n -1

D $\geq 2$

H.W.

#Q. A simple graph with n vertices is necessarily connected if number of edges are more than E, then find the value of E.

H.W.

#Q. Maximum number of edges possible in a simple undirected graph with n-vertices and 'k' components ?

# 2 mins Summary

- Topic: Vertex Covering
- Topic: Minimal & Minimum vertex covering
- Topic: Vertex independent set
- Topic: Maximal & Maximum vertex independent set
- Topic: Connectivity

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# THANK - YOU

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